

The results help us in reviewing our past performance and future planning. After collection of data, the most important step is its presentation.

Tally charts are used for describing, counting, quantity of the grouped data.

Table below represents the data of students of class VII; how they go to school.

How do you go to school?		Number of students
By foot		9
BY car		6
By bus		13

13.1.2 Define frequency distribution (i.e frequency, lower class limit, upper class limit and class-interval).

I. Frequency

The number of values that occurs for a particular group of a data is called its frequency.

Example 1. A group of twenty students of VII class obtained the following marks in a Science test.

16, 11, 40, 27, 38, 55, 45, 70, 50, 65, 60, 41, 100, 71, 80, 75, 82, 85, 92, 89

Steps:

- (i) Arrange the data in ascending order:

11, 16, 27, 38, 40, 41, 45, 50, 55, 60, 65, 71, 70, 75, 80, 82, 85, 89, 92 and 100.

- (ii) Make interval of suitable number of three groups.

First, students who obtained marks from 11 to 40. Second, who obtained the marks from 41 to 70. Third, who secured the marks from 71 to 100.

(iii) Make frequency chart as under:

Name of Group	Period or class interval	Marks obtained	No. of students	Tally mark
First	(11 – 40)	11, 16, 27, 38 and 40	5	
Second	(41 – 70)	41, 45, 50, 55, 60, 65 and 70	7	
Third	(71 – 100)	71, 75, 80, 82, 85, 89, 92 and 100	8	

The frequency chart is as under:

The frequency of (11 – 40) is 5. Here 5 students obtained marks 11 to 40.

The frequency of (41 – 70) is 7. Here 7 students obtained marks 41 to 70.

The frequency of (71 – 100) is 8. Here 8 students obtained marks 71 to 100.

Class Intervals: Each group of a data is also known as the class interval. For example, (11 – 40), (41 – 70) and (71 – 100) are class intervals represent the values of a group.

Upper Class Limit: The greatest value of a class interval is called the upper class limit. For example, in the class interval (41 – 70), 70 is the upper class limit.

Lower Class Limit: The smallest value of a class interval is called the lower class limit. For example, in the class interval (71 – 100), 71 is the lower class limit.

Size of the Class Interval: The number of values in a class interval is called its size or length. For example, the size or length of class interval (11 – 40) is 30.

Example. There are 40 students in the class VII who got the following marks in an English test. Make a frequency table or frequency distribution table by using 5 classes of an equal size.

26, 41, 17, 15, 18, 60, 46, 33, 24, 15, 52, 39, 28, 89, 74, 68, 56, 38, 92, 49, 28, 82, 19, 21, 34, 23, 43, 77, 65, 64, 21, 59, 15, 33, 66, 29, 33, 65, 35, 39.

Solution: Arrange the marks obtained in ascending order as:

9, 15, 15, 15, 17, 18, 19, 21, 21, 23, 24, 26, 28, 28, 29, 33, 33, 33, 34, 35, 38, 39, 39, 43, 46, 46, 49, 52, 56, 59, 60, 64, 65, 65, 66, 68, 74, 77, 82, 89, 92.

We know that,

$$\text{Size of class} = \frac{\text{Greatest value} - \text{Lowest value}}{\text{Number of class intervals}}$$

We can see from the above un-grouped data (arranged in ascending order) that of:

$$\text{Greatest value} = 92$$

$$\text{Lowest value} = 9$$

$$\text{No. of class intervals} = 5$$

$$\text{Therefore size of class interval} = \frac{92 - 9}{5} = \frac{83}{5} = 16.6 \approx 17$$

Class Interval	Tally Marks	Frequency
9 - 25		10
26 - 42		13
43 - 59	I	6
60 - 76	II	7
77 - 93		4

EXERCISE 13.1

1. A book store sold the following number of story books, last week. Construct a frequency distribution table.

Name of Day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Bbooks sold	24	25	27	28	22	30	21

2. The electricity bills paid by 12 consumers are given below. Make a frequency distribution table of 5 classes of an equal size.

Rs 5100, Rs 7000, Rs 3560, Rs 6030, Rs 4220, Rs 6740, Rs 4810, Rs 5450, Rs 7180, Rs 5920, Rs 6850 and Rs 5690.

3. In an annual examination, 20 students of a school got the following marks out of 850 marks. Construct a frequency distribution table by taking 100 as a class interval.

551, 786, 678, 725, 788, 580, 720, 690, 750, 651, 599, 609, 719, 760, 625, 775, 646, 667, 753, 675.

4. The following data shows the number of children who visited the zoo in last month.

134, 167, 145, 130, 155, 142, 130, 180, 162, 130, 120, 165, 170, 110, 200, 185, 132, 140, 110, 170, 100, 160, 133, 150, 125, 110, 145, 155, 160, 200.

Construct a frequency distribution table by taking 10 as a class interval.

5. There are 30 students in class VII who secured following marks in a Mathematics Test.

45, 19, 49, 17, 37, 10, 25, 28, 50, 46, 43, 34, 25, 50, 48, 46, 42, 34, 33, 11, 28, 19, 30, 17, 13, 11, 39, 41, 35, 34.

Make a frequency distribution table using 5 as a class interval of an equal size.

13.2 PIE GRAPH

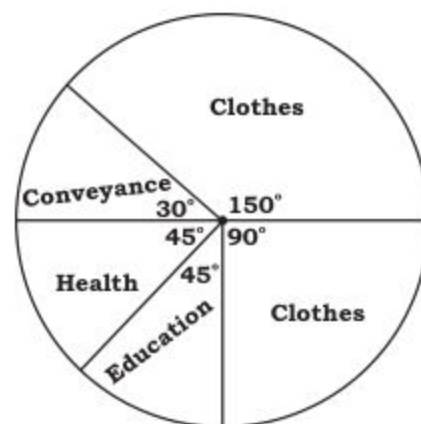
Interpret and draw pie-graph

A pie graph is generally used for the comparison of some numerical facts classified in different classes. In this graph, the central angle measures 360° which is sub-divided into the ratio of the sizes of the groups to be shown through this graph. Following example will help to understand the concept of a pie graph.

Example 1. The expenditure made by a student in a week period is Rs 3600. It is represented through the following pie graph.

Read the pie graph and answer.

- Draw a chart showing the expenditure on each item.
- At what items, he spent most amount? And how much?
- At what item, he spent the least amount? And how much?
- At what items, he spend equal amount? And how much?



Solution: Here whole circle is divided in 360° .

There are five sectors. The biggest sector has an angle of 150° .

It show the expenditure made on food.

$$(i) \text{ Therefore: amount spent on food} = \frac{\text{Rs } 3600 \times 150^\circ}{360^\circ} = \frac{\text{Rs } 3600 \times 150}{360} = \text{Rs } 1500$$

$$(ii) \text{ The amount spent on clothes} = \frac{\text{Rs } 3600 \times 90^\circ}{360^\circ} = \frac{\text{Rs } 3600 \times 90}{360} = \text{Rs } 900$$

$$(iii) \text{ The amount spent on education} = \frac{\text{Rs } 3600 \times 45^\circ}{360^\circ} = \frac{\text{Rs } 3600 \times 45}{360} = \text{Rs } 300$$

Name of item	Food	Clothes	Education	Health	Conveyance
Amount spent	Rs 1500	Rs 900	Rs 450	Rs 450	Rs 300

Pie-graph shows, the most amount is spent on food i.e Rs 1500.

Pie-graph shows, the least amount is spend on conveyance i.e Rs 300.

Pie-graph shows, the equal amount is spent on education and health i.e Rs 450.

Example 2. In District Sports competition of 1800 students 750 played cricket, 200 played badminton, 400 played hockey and 450 played football. Draw a pie graph of the students playing the games.

Solution:

Total number of students = 1800

(i) Find the angle for each sector by using the following formula.

$$\text{Required angle} = \frac{\text{No. of students played a game}}{\text{Total number of students}} \times 360^\circ$$

$$\text{Measure of angle associated with badminton} = \frac{200}{1800} \times 360^\circ = 40^\circ$$

$$\text{Measure of angle associated with cricket} = \frac{750}{1800} \times 360^\circ = 150^\circ$$

$$\text{Measure of angle associated with hockey} = \frac{400}{1800} \times 360^\circ = 80^\circ$$

$$\text{Measure of angle associated with football} = \frac{450}{1800} \times 360^\circ = 90^\circ$$

(ii) To draw a pie graph.

Step 1: Draw a circle of suitable radius.

Step 2: Draw an angle of 40° representing the badminton: (40°)

Step 3: Draw an angle of 150° representing the cricket: (150°)

Step 4: Draw an angle of 80° representing the hockey: (80°)

Step 5: Remaining angle representing the football will be of: (90°)

Total (360°)

(iii) Label each sector according to the figure.

(iv) Answer the following:

1. Which is the most favourite game?

Cricket.

2. Which game is liked by the least number of students?

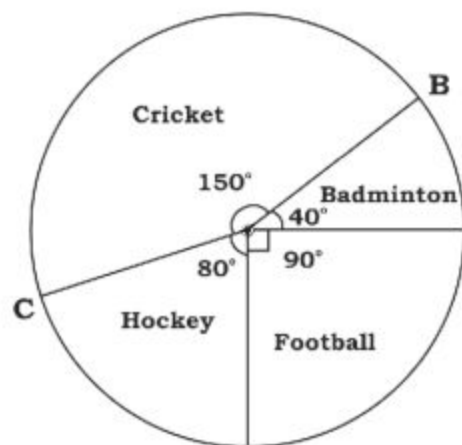
Badminton

3. How many more students played football than hockey?

50 students

4. How many less number of students played badminton than hockey?

200 students



EXERCISE 13.2

- In mathematics quiz 80 students appeared. 60 students passed and 20 students failed. Represent it through pie chart.
- In class VII there are 36 students. The favourite colour of 16 is red, 9 is blue, 7 students like pink colour and four liked white. Make a pie chart of it.
- Sara and her friends eat fruits in a week as shown in the table.

Name of fruit	Apple	Banana	Mango	Peach
No. of fruits	12	24	16	8

By using the table draw a pie graph.

4. In a party, Mr. Jamal served the guest by following food items.

Food Items	Cold drink	Sandwich	Burger	Samosa
Quantity	180	124	330	86

Use the table to draw a pie graph.

5. The following data shows the distance in km that Hooria travelled in last month.

90, 44, 55, 49, 28, 9, 92, 27, 18, 84, 50, 60, 79, 69, 24, 89, 63, 74, 35, 48, 39, 80.

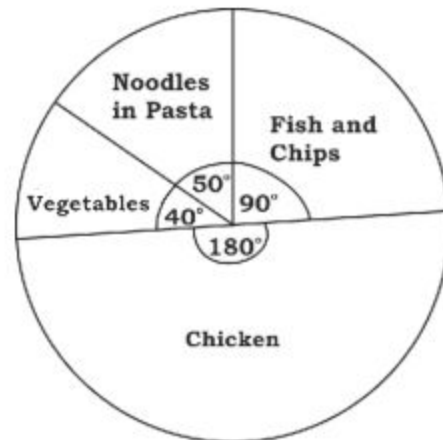
Construct the frequency table using 6 as class interval of an equal size. Also draw its pie chart.

6. Hina asked 180 students, about their favourite meal.

She drew a pie graph.

Read the pie-graph and answer the following:

- What is the most favourite meal? And how many students liked it?
- Which food is liked by the least number of students? And how many students liked it?
- How many students liked noodles in pasta?
- Use the pie graph and complete the table.



Favourite meals	Frequency	Angle
Chicken	90	180°
Noodles in pasta		50°
Vegetables		
Fish and Chips	40	
Total	180	

REVIEW EXERCISE 13**1. Fill in the blanks.**

- (i) Information are called _____ which are normally the results of measurements, observations or experiments.
- (ii) The number of values that occurs in a group of data is called _____.
- (iii) A data is collected in _____ form and it provides us information about individuals.
- (iv) The smallest value of a class interval is called _____.
- (v) The table which shows the frequency of class intervals is called _____.
- (vi) After collection of data, the most important step is _____.
- (vii) The method which is used to record the result is called _____.
- (viii) The greatest value of a class interval is called the _____ limit.
- (ix) The number of values in a class interval is called its _____.
- (x) The representation of a numerical data in the form of sectors of a circle is called a _____.

2. Tick (✓) the correct answers.

- (i) In pie graph, the measure of central angle is:
(a) 90° (b) 18° (c) 24° (d) 360°
- (ii) In the class interval (10 – 50), the upper class limit is:
(a) 11 (b) 10 (c) 50 (d) 51
- (iii) In the class interval (9 – 12), the lower class limit is:
(a) 9 (b) 7 (c) 13 (d) 14
- (iv) In a grouped data, greatest value = 21, lowest value = 3, and number of interval = 3, then the size of the class is:
(a) 3 (b) 6 (c) 18 (d) 21

3. The ages of visitor in park during a week are given below. Group the data taking 10 as the size of an interval.

25, 50, 49, 47, 26, 10, 2, 1, 15, 17, 18, 19, 27, 28, 30, 35, 17, 32, 31, 3, 4, 9, 10, 15, 12, 13, 17, 24, 20, 22, 24, 26, 30, 17, 35, 40, 36, 32, 31, 37.

SUMMARY

1. Data means groups of information that are normally the results of measurements, observations experiments.
2. Data is collected in raw form and it provides us information about individuals. Such form of the data is called ungrouped data.
3. The number of values that occurs in a class interval is called its frequency.
4. In a grouped data, each group is also known as the class interval.
5. The greatest value of a class interval is called the upper class limit.
6. The smallest value of a class interval is called the lower class limit.
7. The table which shows the frequency of class intervals is called frequency table.
8. The representation of a numerical data in the form of disjoint sectors of a circle is called a pie graph.
9. In pie graph, the central angle measures 360° which is divided into the ratio of the sizes of the groups.

Think Tank: The Most Wonderful Knowledge.

A teacher asked the students to simplify $\frac{16}{64}$ to its lowest term. Abid cancelled 6 in the numerator with 6 in the denominator and brought the answer $\frac{1}{4}$. The answer is no doubt correct! Should the teacher give him marks for solving the example? Not at all. Again try with $\frac{19}{95}$. The answer is correct and yet it does not merit marks. Similarly try with other numbers also.

GLOSSARY

Additive inverse:	Two rational numbers whose sum is equal to zero, are additive inverses of each other ($-\frac{2}{3}$ is additive inverse of $\frac{2}{3}$).
Adjacent angles:	Two angles that have a common vertex, a common arm and no. point common in their interiors.
Associative property of union of sets:	$A \cup (B \cap C) = (A \cup B) \cap C$
Associative property of intersection of sets:	$A \cap (B \cup C) = (A \cap B) \cup C$
Algebraic expression:	It is a combination of constants and variables connected by the signs of fundamental operations (+, -, ×, ÷).
Arc:	A part of the circle.
Area of circle:	Area of circle = πr^2
Base:	When a number is written in exponential form, the number which is to be multiplied by itself according to exponent. e.g. In $2^3 = 8$, the base is 2.
Binomial:	A polynomial having two terms. e.g. $x + y$, $3a - bc^2$ and $y^2 + z$.
Complement of set A:	It is a set of all members of U which do not belong to A and is denoted by A' . ($A' = U - A$).
Commutative property of union of sets	$A \cup B = B \cup A$.
Commutative property of intersection of sets:	$A \cap B = B \cap A$.
Circumference:	It is the length of the boundary of a circle.
Complementary angles:	Two angles are complementary when their sum is 90° .
Continued ratio of three quantities:	It is the combination of two ratios with at least one common quantity with same size:
Class interval:	In grouped data, each group is class interval.
Descriptive form:	A form of a set which describes characteristics or properties of the members of the set in any common language e.g. A = set of days of a week.
Difference of two sets:	Difference of A and B, denoted as $A - B$, is the set which contains all the members of A which are not in B. $A - B \neq B - A$. $A - B$ is also represented by $A \setminus B$.
Disjoint sets:	Those sets which have no common member. e.g. $A = \{1, 2, 3\}$ and $B = \{p, q, r\}$ are disjoint sets.
Direct proportion:	A relation in which increase or decrease in one quantity causes corresponding increase or decrease in the other.
Decimals:	It is a number with decimal point.
Exponent:	It is a number which represents the number of times; a base is to be multiplied by itself.
Frequency:	It is the number of values that occur for a particular group of data. e.g. The frequency of (11 - 40) is 5. It means only 5 values are between 11 to 40.
General sales tax (GST):	It is a tax paid to the Government for sales of certain goods and services. The rate is 17% since 2014.
Hypotenuse:	In right angled triangle, it is the longest side opposite to right angle.